103. Fat present below the skin surface in our body acts as a barrier against
(a) Loss of heat from body
(b) Loss of essential body fluids
(c) Loss of salt from body
(d) Entry of harmful micro-organisms from the environment.

Ans. (a): Fat present below the skin surface in our body, acts as a insulator against loss of heat from body. The deposited fat works a barrier, which prevents loss of heat.

104. Which of the following professional(s) are more likely to run the risk of permanent change in their cell's DNA?
1. Researchers using carbon 14 isotope
2. X-ray technician
3. Coal miner
4. Dyer and Painter

Select the correct answer by using the codes given below
Codes:
(a) 2 alone (b) 1, 2 and 3
(c) 1, 2 and 4 (d) 1, 3 and 4

Ans. (a): X-Ray technique, C^{14} and Dyer technique is very important technique for permanent change in DNA. But in coal miner C^{14} amount is very less so, there is no risk of change in DNA.

105. Of the four land marks in medical history given below, which one was the first to take place?
(a) Organ Transplant (b) Bypass surgery
(c) Test Tube Baby (d) Plastic surgery

Ans. (d): Organ Transplant – 1950
Bypass surgery – 1960
Test Tube Baby – 1978
Plastic Surgery – 1827

106. Match List-I with List-II and the correct answer by using the codes given below:

List-I	List-II
A. Vitamin	1. Pepsin
B. Enzyme	2. Carotene
C. Hormone	3. Keratin
D. Protein	4. Progesterone

Codes:
(a) A-1; B-2; C-3; D-4 (b) A-2; B-1; C-4; D-3
(c) A-2; B-1; C-4; D-4 (d) A-1; B-2; C-4; D-3

Ans. (b): Vitamin A group includes retinal. Their parent carotene is a vitamin substance as B-carotene.
Pepsin is an enzyme produced in stomach, which digest protein.
Progesterone is a hormone, produced in human ovary
Keratin is a protein. The main component of hair and nail is keratin.

107. Besides Proteins and carbohydrates, other elements of nutritional value found in milk, include
(a) calcium, potassium and iron
(b) calcium and potassium
(c) potassium and iron
(d) calcium and iron

Ans. (b): Besides carbohydrates an protein calcium, magnesium an phosphorus is found in milk.

108. Match List-I (Physiological process) with List-II (cell organelles) and select the correct answer by using the codes given below:

List-I	List-II
A. Photosynthesis	1. Plasma membrane
B. Mineral uptake	2. Chloroplast
C. Respiration	3. Mitochondria
D. Protein synthesis	4. Ribosomes

Codes:
(a) A-1; B-2; C-3; D-4 (b) A-2; B-2; C-4; D-3
(c) A-2; B-1; C-3; D-4 (d) A-2; B-1; C-4; D-3

Ans. (c): Photosynthesis occurs in green plants by help of chloroplast plasma membrane helps in transport of minerals cellular respiration is related to mitochondria, by which energy releases. Ribosome helps in protein synthesis.

109. Which one of the following is a modified stem?
(a) Carrot (b) Sweet potato
(c) Coconut (d) Potato

Ans. (d): Potato tubers bears buds in small pits known as eyes. Buds develops to branches some of the branches become green, erect an leafy stems that grow horizontally underground.

110. Which one of the following is not an essential micronutrient for plants?
(a) Boron (b) Zinc
(c) Sodium (d) Copper

Ans. (c): There are 13 essential nutrients required by plants for its healthy an proper growth. Now these nutrients are divided into two categories:
Macronutrients (iron, copper manganese, zinc, boron, molybdenum and chloride) Additional mineral nutrient elements which are beneficial but not necessary are sodium, cobalt, vanadium, nickel, selenium, aluminum and silicon.
Thus boron, zinc and copper falls into category of essential macronutrients while sodium does not.

111. The nutritional deficiency condition the needs to be given top priority for remedial action in India today is
(a) Scurvy (b) Rickets
(c) Xerophthalmia (d) Pellagra

Ans. (c): Xerophthalmia is caused by deficiency of vitamin A. Generally this disease is found in poorer section of society. The symptom includes high blindness and eye irritation.

ENVIRONMENT AND ECOLOGY

112. Which one of the following regions of India is now regarded as an 'Ecological hot spot'?
(a) Western Himalayas (b) Eastern Himalayas
(c) Western Ghats (d) Eastern Ghats

Ans. (c): Biodiversity hotspots in India includes Western Ghats, the Himalayas, Indo-Burma border and Nicobar Islands (Sundaland hotspot). Hence option (c) is correct.

113. Given below are two statements, one labeled as Assertion (A) and other labeled as Reason (R):
Assertion (A) : Mangroves are very specialized forest ecosystems of tropical and sub-tropical regions bordering certain sea coasts
Reason (R) : They stablise the shoreline and act as bulwark against encroachments by sea.

In the context of above two statements, which one of the following is correct?
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (a): Mangroves are special type of vegetation and are found in intertidal regions where freshwater and saltwater intermixes in the bays, estuaries, creeps and lagoons. They are found in tropical and subtropical regions. They stablise shoreline and act as bulwark against encroachments by sea.

114. Consider the following programmer:
1. Afforestation and development of Wastelands
2. Reforestation and replantation in existing forests
3. Encouraging the wood substitutes and supplying other types of fuel.
4. Promotion of wide use of insecticides and pesticides to restrict the loss of forest area from degradation caused by pests and insects.

It can also effect central nervous system and liver.
The National forest policy of 1988 includes:
(a) 1, 2, 3 and 4 (b) 2 and 4
(c) 1, 3 and 4 (d) 1, 2 and 3

Ans. (d): National Forest Policy, 1988 describes the protection of forest. It does not describe the use of insecticide and pesticide in forest areas.

CURRENT AFFAIRS

115. The following table shows the percentage distribution of revenue expenditure of Government of India in 1989-90 and 1994-99

Expenditure Heat (Percentage total)					
Years	Defence	Interest	Subsidies	State/UTs Grands	
		Payments		to others	
1989-90	15:1	27.7	16.3	13.6	27.4
1994-95	13.6	38.7	8.0	16.7	23.0

Based on this table, it can be said that the Indian economy is in poor shape because the central government continues to be under pressure to:
(a) reduce expenditure on defence
(b) spend more and more on interest payments
(c) reduce expenditure on subsidies
(d) spend more and more as grants-in-aid to state government/Union Territories.

Ans. (b): According to the data showing revenues expenditure of Indian government, it can be said that the Indian economy is in poor shape due to it's increasing dept payments due to burden of debt payments there remains little scope to utilize this money for any other purpose.

116. Which one of the following statements is to be launched from India in 1996
(a) IRS-P2 (b) IRS-P3
(c) IRS-EN (d) IRS-ID

Ans. (b): IRS-P3 was an experimental earth observation mission undertaken by ISRO. The objectives of the mission were processing and interpretation of data generated by its two payloads, the wide field sensor and Modular Opt electric sensor, development by the German Aerospace centre. The mission was completed during January 2006 after serving for a years 10 months.

117. Consider the following statements:
Most international agencies which Fund Development Programme in India on intergovernmental bilateral agreements mainly include:
1. Technical assistance
2. Soft loans which are required to be paid back with interest
3. Grants, not required to be paid back
4. Food assistance to alleviate poverty

(a) 2 and 4 are correct
(b) 1, 2 and 3 are correct
(c) 1, 2 and 4 are correct
(d) 3 and 4 are correct